Nexus AI OS: Autonomous Edge Intelligence Framework
Department of Computer Science & Engineering - Capstone Project

Abstract:
This research document demonstrates the feasibility of combining lightweight local Large Language Models (LLMs

Key Findings:
1. Low Latency: 4B parameter models running via Ollama provide under-2-second inference on standard 16GB RA
2. Safe Filesystem Operations: Strictly bounded path validation prevents traversal attempts (../) while allowing rapi
3. System Telemetry: Hardware metrics acquired via psutil enable responsive OS dashboards without blocking infe
4. Privacy: All user interactions remain on-device with zero cloud telemetry.